Natural language Processing

A.V. Ameya
192472036
05A0208

CIA-1

1. Regular expressions

Regular expressions are the part of natural language processing is used to identify the text from the information.

ex hospital system stores a clinical as plain text by using phone number, date, email etc....

They are four lines to be processed

1. Date
2. Number
3. Email

1. Date

^ [1-3] \ [1-1] {1,2} \ $

→ Days must be between 1-31

→ month must be between 1-12

→ $ end of the input

2. Phone number

^ [6-9] [0-9] {9} \ ^

→ The phone number should contain 10 digits

→ The first digit should be between 6 or 9.

→ The remaining digit should between 0-9.

3. Email

{}{3 dd. gmail @ com

→ The mail id should start with the user name.
→ The mail should contain the numbers.
→ The mail have the symbol '@.'

The 4 lines to be processed

b1 05/03/2025 – Valid
b2 98 76543210 – Valid
b3 5876543210 – Invalid
b4 5/3/26 – Invalid

→ b3 is invalid because the starting number should start b/w 6 & 9.

→ b4 is rejected because with no proper zeros.

Conclusion

Therefore regular expression helps us to understand the words from the text, which makes the information easier.

2. Finite state automato

The finite sate automato is the part of the NLP is used to understand the patterns from the text.

They are three types

1. hexion level
2. Intermidate level
3. Finial level

hexion level

It is the orginal form of the word

Ex:- fox, cat, dog

Intermidate level

It is the bansanformation of a word

goes fox + es

cats + s

Finial level

It is finial output of the word

Ex:- foxes, cats.

State	bansatitions
fox	accepted
cat	accepted
dog	accepted
goose	accepted
mouse	rejeted
geese	accepted
mice	rjejected → already pural

Morpheme sequence

State	bansactions
dogs	accepted
mouse	rejected → irregular noun
geese	accepted
foxs	accepted
mices	rejected → already plural

Morphological

Dividing the word into the simpiler form

walked	infectional
unhappniess	derviational
computrijation	infectional
geese	infectional

3. Porter stemmer

Porter stemmer is an algotherim is used to divide the ~~word~~ text into simplier understamble word form.

Steps

1. By removing the plural forms in a word
2. By adding the saffin to the word
3. By removing the additional suffin to the word
4. Analuzing the word
5. Generating the word

Name	step	stage
ponies	poni+es	poni
motoring	motor+ing	motor
releational	releat+ional	releat

hopefulness	hope+fulness	hope
generlization	general+ization	general

b.

N-gram model

The N-gram model is the predecting of the next word in the sentence.

They are two types.

a. Unigram model

eg Unigram model is the probality of the particular word.

b. bi-gram model

eg bi-gram model is the probality of the word depends on the front word.

S_1 = "the cat sat"

S_2 = "the dog sat"

S_3 = "the dog ran"

Unigram

the = 3

cat = 1

dog = 2

Sat = 1

ran = 1

Bigram

dog = 2

the dog = 1

dog sat = 1

dog ran = 1

Cat sat = 1

- N- gram model

N-gram model is the probality of find the word in the sentence.

They are too types....

a. Unigram model

It is the probality of the particular word in a sentence.

b. Bigram model

It is the probality of the coord which depends on the front coord

< I like tea >

< I like coffee >

< I drink tea >

Unigram

I = 3

like = 2

tea = 2

Coffee = 1

drink = 1

Bigram

tea = 2

like tea = 2

drank tea = 2

like tea = 1

like coffee = 1

I like = 2

I drink = 1

Probality = N x d x λ

= 12 x 0.5 x 0.7

= 4.2

5. Pos tagger

Part of speech tagger is the part of NLP is used to identify the word tense.

They are .... PRP, MD, VB, NN

" They can fish "

Harkov marking model (HMM)

It is the part of the POS tagger.

1. Transition HMM
2. Emission HMM

$V_B = 0.4 \wedge 0.5 \wedge 0.02 = 0.004$ max

$= (0.04 \times 0.8, 0.004 \times 0.1, 0.004 \times 0.4) \times 0.01$

$= 0.0384 \times 0.01$

$= 0.000384$

$NN = 0.4 \times 0.1 \times 0.001 = 0.0004$ max

$= 0.0096 \times 0.02$

$= 0.000192$

Best finial State VB. Bank having gives PRP on with 3.84×10^{-11} probality

Rule 1 :- The word it can be candiate tags & the perviows words in proper pronoun

eliminate NN & VD

So MD = Can

Rule 2 :- The previous word tag is MD. Estimate NN fish = VB. So the constants rule progressive set until the tag is set.

Transaction bak Kad tagging

The PP and NN with 1 error out of 3 tokens. The learn bies the templates over the previous tags.

Comparsion

Approach knowledge strength weakness. Rule base hand-written constraints is a lexican. Transits high prsicion to write, hard to maintain Stochasic (HMM) tag & word probalities from a corpus. learns automatically ranks alternative. Need a tagged corpus,